The Future of capitalism

Lecture 2: The Finance view

Mon 9 Feb 2026, 15.30

HG 11A-33

VU VRIJE UNIVERSITEIT AMSTERDAM

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Amartya Sen (1933)

“We live in a world of unprecedented opulence, of a kind that would have been hard even to imagine a century or two ago.”

“And yet we also live in a world with remarkable deprivation, destitution and oppression.”

Sen, Amartya (1999). Development as Freedom (Function). Kindle Edition.

Agenda for today

1. Testing Marx (brief, for self-study. See teaching note 2a)
2. The Finance view: why growth? What is the role of corporate finance & investment?

Marx applied to the workplace: wage gap and child penalty

- Men and women have different power positions in society, because of norms and values “common sense” that have been inherited from earlier times.
  - Power dynamics: women suffer financially because men dominate in the workplace. They increase their human capital while women stay at home, so they “own the means of production” (Foucault)
  - Bourgeois attitudes: women are expected to take care of children. (Gramsci)
- We could test this by looking at the wage gap:
  - Do men earn more than women?
  - Do earnings change after birth of a child (*proles*)?

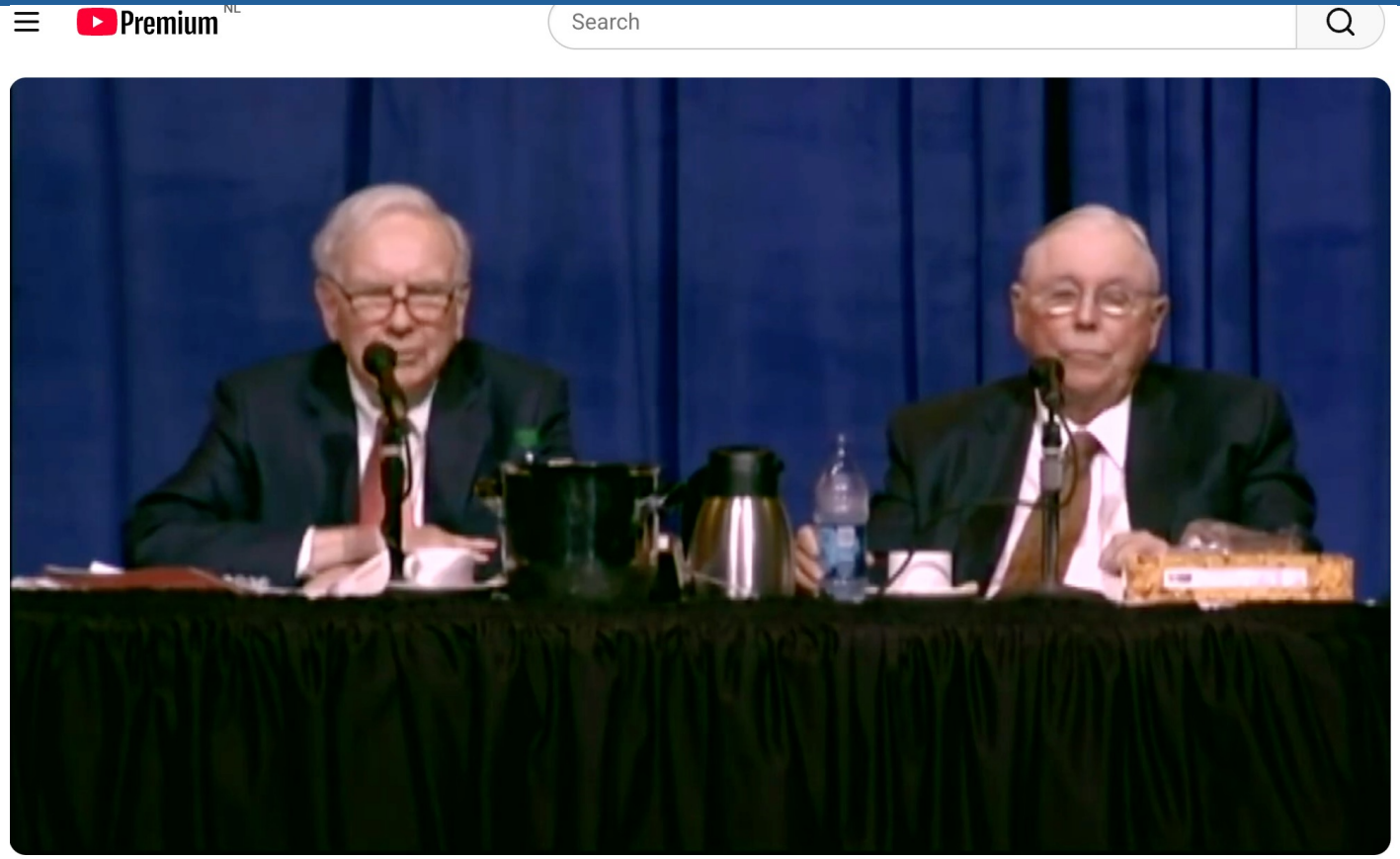
I. Testing the wage gap

- We want to measure the wages difference between men and women
- How would you test that empirically? Say, on a dataset of men and women. Individual i, time t
  - $Wage_i = a \cdot gender_i$?
  - $Wage_i = a \cdot gender_i +$ ...?

II. The Child penalty

- We want to measure the relative change in wages occuring after the birth of a child, separate for men and women
- How would you test that empirically? Say, on a dataset of men and women. Individual i, time t
  - $Wage_i = a \cdot gender_i$?
  - $Wage_i = a \cdot gender_i \times I_{recent\ birth}$?

What to do in a tough Job market



Buffett & Munger: What To Do In A Tough Job Market

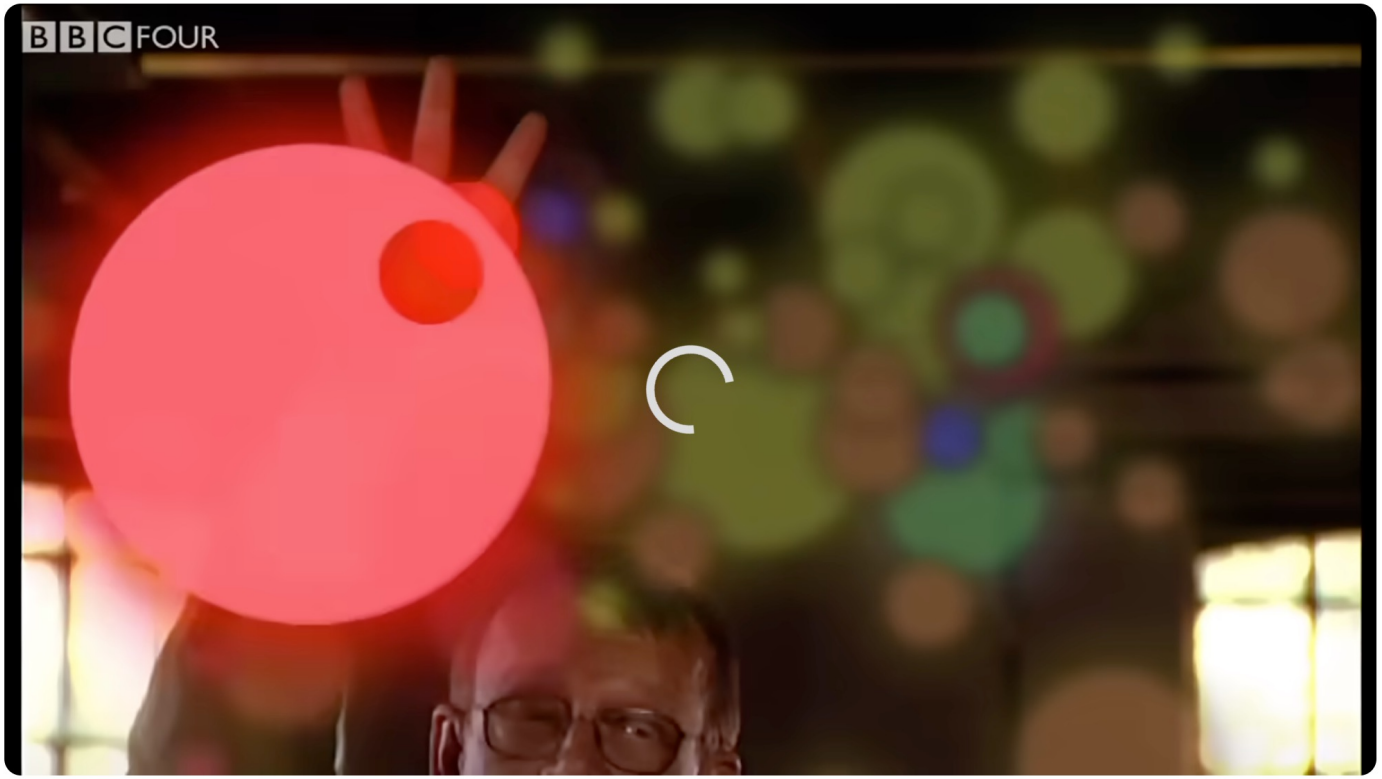
Epilogue: Letter of Proudhon to Marx (p153)

Let us together seek, if you wish, the laws of society, the manner in which these laws are reached, the process by which we shall succeed in discovering them; but, for God's sake, after having demolished all the *a priori* dogmatisms, do not let us in our turn dream of indoctrinating the people. . . . I applaud with all my heart your thought of inviting all shades of opinion; let us carry on a good and loyal polemic, let us give the world the example of an informed and farsighted tolerance, but let us not—simply because we are at the head of a movement—make ourselves into the leaders of a new intolerance, let us not pose as the apostles of a new religion, even if it be the religion of logic, the religion of reason. Let us gather together and encourage all dissent, let us outlaw all exclusiveness, all mysticism, let us never regard a question as exhausted, and when we have used one last argument, let us if necessary begin again—with eloquence and irony. On these conditions, I will gladly enter into your association. Otherwise, no!

The letter is symbolic:

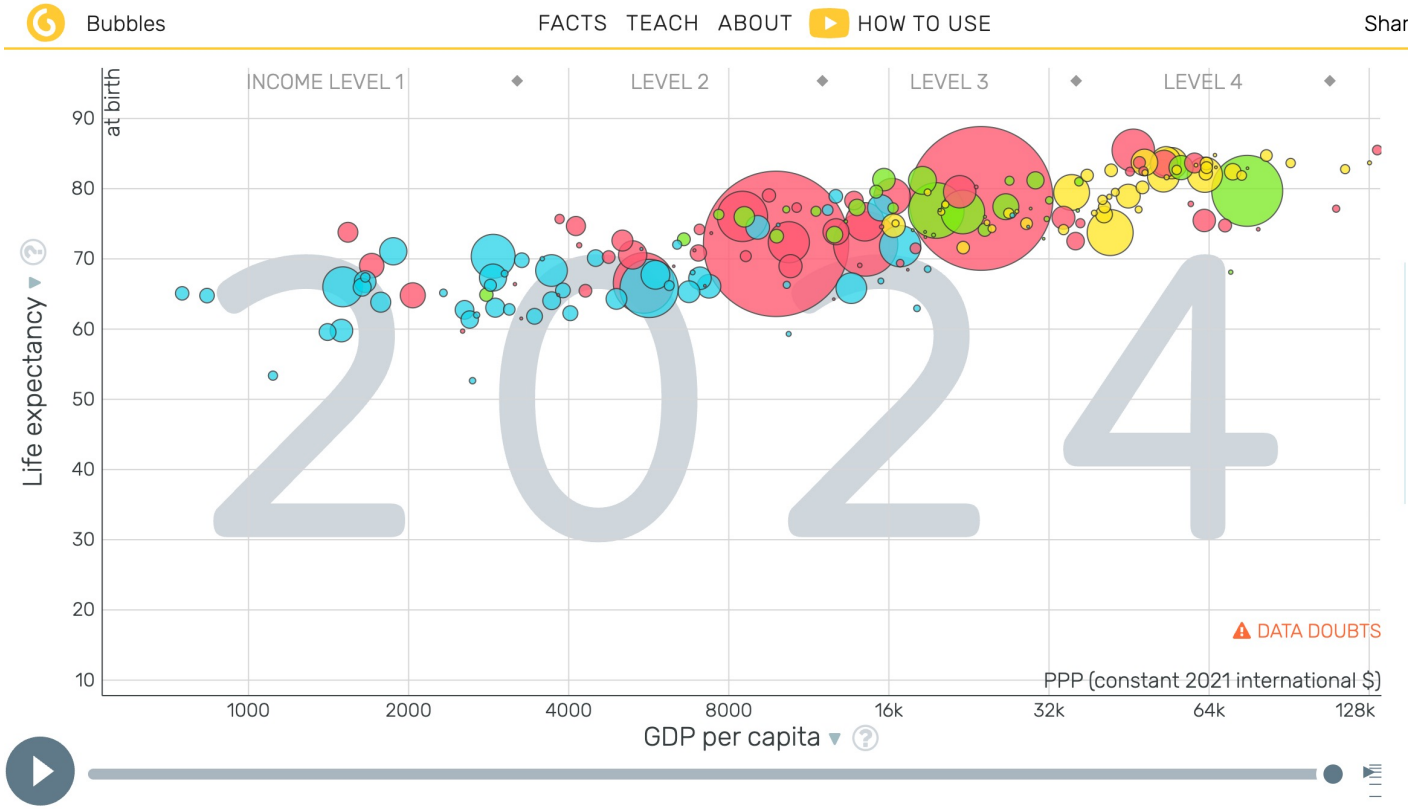
- "The peculiar tone which Marx injected into working-class affairs" (152)
- "The pattern of intolerance was never to disappear" (154)
- "that infuriating and absolute inability to entertain dissent, which communism has inherited from its single greatest founder" (154)

The surprise of human progress (1870--)



Hans Rosling's 200 Countries, 200 Years, 4 Minutes - The Joy of Stats - BBC

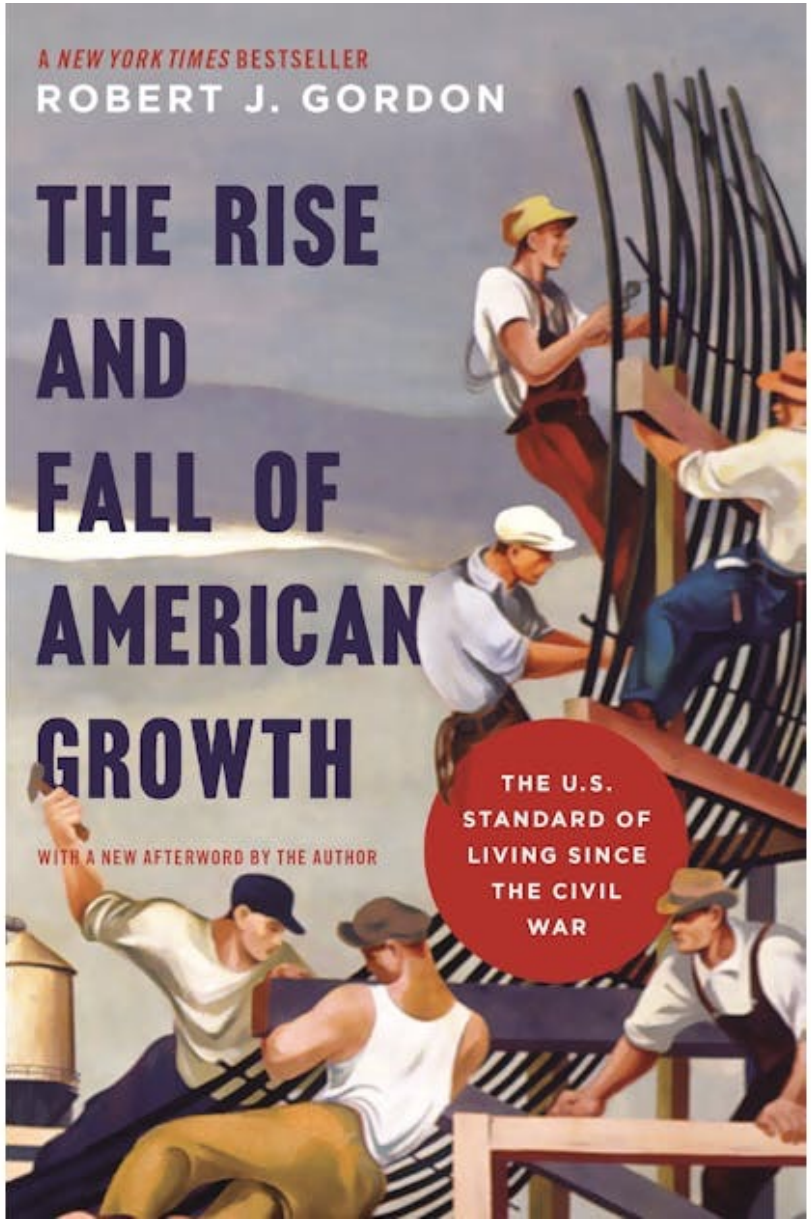
2024


Bubbles
FACTS
TEACH
ABOUT
HOW TO USE
Shar
INCOME LEVEL 1
LEVEL 2
LEVEL 3
LEVEL 4
Life expectancy at birth
90
80
70
60
50
40
30
20
10
2024
DATA DOUBTS
PPP (constant 2021 international $)
1000
2000
4000
8000
16k
32k
64k
128k
GDP per capita

The reality of modern economies is in stark contrast with Marx' predictions

- A large middle class => couldn't keep existing
- Relatively low inequality and poverty (compared to the 1830s!)
- Employment with high wages => no proletarians
- Economic growth => continuing innovation
- Many small and medium-sized entrepreneurs => no takeover by large firms and rent-seeking capital owners

- … and a very large financial sector.
  - How does Finance create real economic value?
  - And will growth continue? How do we know future growth is going to be >0?


A NEW YORK TIMES BESTSELLER
ROBERT J. GORDON
THE RISE AND FALL OF AMERICAN GROWTH
WITH A NEW AFTERWORD BY THE AUTHOR
THE U.S. STANDARD OF LIVING SINCE THE CIVIL WAR

Marx gave the financial sector only secondary attention

- Interest-bearing capital: a **“fetish form”** of capital: money seems to produce more money without exploitation or production being visible
- Capital separated from ownership can be “parasitic” on surplus value produced elsewhere

- In all, the financial sector itself did not play a large role in Marx’ reasoning. The same holds, by the way, for macro-economists until 2008…
  - Many in history have seen finance (banks) useless or evil. The “money lenders”. Aristotle. Association with anti-semitism. “the power of money” => “it is the Jews”. Read Shakespeare’s merchant of Venice if you like.
- In modern economies, the sector is large and continues to increase.

The Big Question

> *"In my field, this is all we do, but you Finance people never seem to care."*
> — Jos, colleague from spatial economics, on studying welfare

How does Finance benefit society?

How can we mitigate its downsides? And what does Finance contribute to growth and prosperity? And can growth continue indefinetely?!

Four Pillars of the Finance View

1

Value creation

NPV

Capital budgeting drives exploration

2

The Power of Risk

Equity as an Option

Limited liability enables exploration and innovation

3

The Power of Diversification

CAPM

How investors can fund risky ventures at reasonable rates

4

The Power of Banks

Diamond-Dybvig

How banks transform illiquid assets into liquid savings

A balance sheet

Assets (A)	**Equity (E)**
	Debt (D)

Part 1

Value creation

NPV computations

Managers create value when they do capital budgeting: select projects with the highest NPV.

A balance sheet

Say we are given an investment opportunity of US$ 100 mln, with a cash flow of US$ 121 at the end of year 1. With a discount rate of r, we would value this project as follows:

$$NPV = -100 + 121/(1 + r) \tag{1}$$

- How is value created? And when?

Capital budgeting as value creation tool

- As long as managers are disciplined in how they deploy capital (capital budgeting), they will select projects that create value.
- Firms should aim for returns on investment that beat the cost of capital (say, a hurdle rate)

- NB: The selection of discount rates matters for the direction of growth and innovation. Let's look at what Shell does

Discount rates for Shell

2021

			GROWTH PILLAR: THE FUTURE OF ENERGY		TRANSITION PILLAR: ENABLING OUR STRATEGY		UPSTREAM PILLAR: FUNDING OUR STRATEGY
		RDS	Marketing	Renewables and Energy Solutions	Integrated Gas	Chemicals and Products	Upstream
Cash capex	Net debt >$65 bilion	$19-22 billion	~$3 billion	$2-3 billion	~$4 billion	$4-5 billion	~$8 billion
			~25%		35-40%		35-40%
	Net debt <$65 bilion	$23-27 billion	~30%		35-40%		30-35%
	Beyond 2025		35-40%		30-40%		25-30%
Average project returns (IRR)			15-25%	Integrated Power IRR >10%	14-18%	10-15%	20-25%
Hurdle rate (IRR)			15%	Integrated Power IRR >10%	12%	12%	18%
Payback period			4-8 years		before 2040	~10 years	before 2035

2023

15%	6-10%	11%	12%	15%

R&ES excl. power 10%
Power generation 6-8%

Part 2

Limited Liability

The innovation of the firm

The corporation as a unique structure that enables exploration and innovation

The Corporation: A Legal Innovation

The Dutch East India Company (VOC), 1602

Transferable Shares

Capital pooled from many investors without personal relationships

Limited Liability

Investors lose at most their investment—personal assets protected

Separate Legal Personality

Entity distinct from owners; can own property, enter contracts, sue

Separation of Ownership & Management

Professional managers run operations; investors provide capital

Equity as a Call Option

Limited liability is mathematically equivalent to a call option.

$$C = \max(S - K, 0)$$

Call Option Payoff

S = Asset value at maturity
K = Strike price (debt level)
C = Equity value

Example: Firm with $20M assets, $10M debt, $10M equity

If assets = $20M → Debt repaid ($10M), Equity gets $10M
If assets = $15M → Debt repaid ($10M), Equity gets $5M
If assets = $8M → Creditors get $8M (loss), Equity gets $0
If assets = $0 → Both worthless, but equity holders lose only their $10M investment

Incentives Under Limited Liability

Equity holders like volatility

Downside capped at investment; upside unlimited. Higher leverage makes risky bets more attractive.

Equity holders like exploration

New markets and innovations are high-variance. Limited liability makes bold ventures fundable.

Delegation to entrepreneurs

Separation of ownership and management allows specialized expertise and efficient decisions.

Transferable shares increase dynamism

Investors not locked in; can sell to optimists. Disciplines management, facilitates reallocation.

Side note: The Solow Residual

The Cobb-Douglas production function:

$$Y = K^{a} \cdot L^{1-a}$$

Y = size of the economy (GDP)
K = capital
L = labor

We can imagine how economies grow by applying more capital and labor.

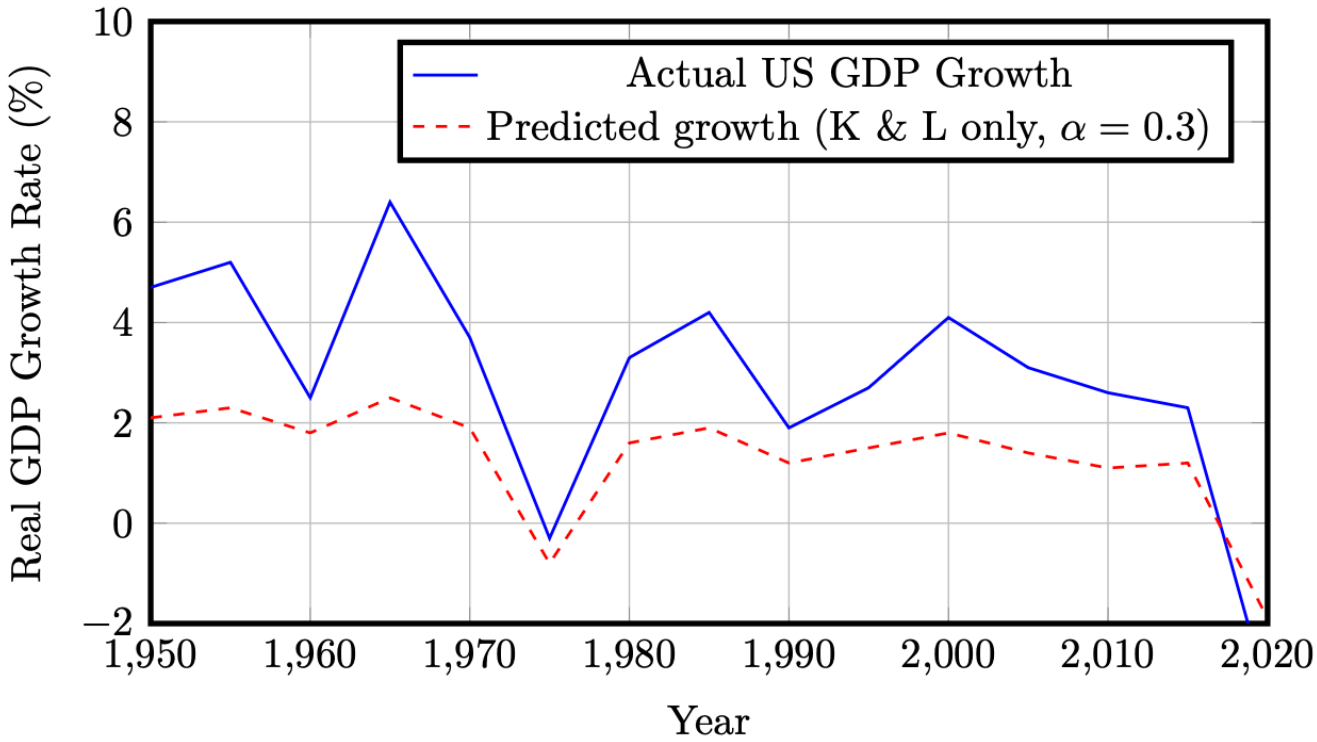


Figure 4: US GDP Growth: Actual vs. Predicted by Capital and Labor. The gap is the Solow Residual (TFP growth).

The persistent gap shows that something besides capital and labor is driving growth. That

Side note: The Solow Residual

87%

of growth unexplained
by capital & labor

The Cobb-Douglas production function:

$$Y = A \cdot K^{a} \cdot L^{1-a}$$

Where A is Total Factor Productivity—the "residual" that captures ideas, techniques, and innovation.

Bottom line:

You cannot create a sustained incentive to generate the Solow residual without some way to reward innovation and risk-taking. The limited-liability corporation, combined with equity markets and the profit motive, provides exactly that incentive.

Part 3

The Power of Diversification

CAPM and the Cost of Capital

Risk doesn't disappear under limited liability—it shifts.
How can risky ventures get funded at reasonable rates?

The SpaceX Paradox

> *"SpaceX has...ready access to cheap capital, thanks to the high valuation investors have been prepared to put on its business."*
> — Financial Times, 2021

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The Puzzle

Rockets explode. Development timelines slip. Competition is intense. Yet SpaceX raises capital at valuations implying low cost of capital.

The Answer: Diversification

Venture capital investors diversify across portfolios. Much of SpaceX's volatility is idiosyncratic and cancels out. Investors care about beta, not total volatility.

CAPM: The Core Insight

$$r_i = r_f + \beta_i(r_m - r_f)$$

Capital Asset Pricing Model

r_i	Required return (cost of capital) for asset i
r_f	Risk-free rate (government bond)
β_i	Beta: sensitivity to market returns
$(r_m - r_f)$	Market risk premium

Cost of capital depends only on systematic risk (β), not total volatility. Idiosyncratic risk can be diversified away for free.

Part 4

The Power of Banks

Diamond-Dybvig and Financial Regulation

Savers want liquidity. Investments need long-term commitment.
How is this mismatch bridged?

Banks: Maturity Transformation

Banks borrow short (deposits) and lend long (loans)

Assets	Liabilities
Risky Loans: $100	Deposits: $90
	Equity: $10
Total: $100	**Total: $100**

The Problem: Liquidity Shocks

If too many savers withdraw at once, the bank must liquidate assets at a discount. If withdrawals exceed ~$50M, losses exceed equity → insolvency.

The Diamond-Dybvig Problem

Self-Fulfilling Bank Runs

Once savers become aware of fragility, they have an incentive to withdraw early. If you believe the bank will fail, you want to be first in line. This creates a bank run: savers rush to withdraw simultaneously. Even a healthy bank can fail if many withdraw at once.

This fragility is not a flaw in bank management. It is a structural feature of maturity transformation itself.

Diamond & Dybvig (1983)

The Regulatory Solution

Deposit Insurance

Government guarantees deposits (e.g., FDIC). Removes incentive to run.

Capital Requirements

Minimum equity-to-assets ratio (8-10%). Cushion to absorb losses.

Liquidity Requirements

Minimum liquid assets vs. deposits. Meet normal withdrawals.

Supervisory Oversight

Regulators monitor risk exposures. Force corrective action early.

The Irony: Without regulation, banks cannot function well. With regulation, they enable growth.

The Finance View: Summary

1. Limited liability enables risk-taking and innovation through the option-like payoff structure of equity.
2. Limited liability enables risk-taking and innovation through the option-like payoff structure of equity.
3. Diversification allows risky ventures to find funding at reasonable rates—only systematic risk (beta) is priced.
4. Banks provide liquidity and maturity transformation, but require regulation to prevent self-fulfilling runs.

Finance theory explains what Marx could not: a mechanism for sustained innovation and growth.

Afterthought: Why no European Google?

> "if we changed how we measure success, Europe would be incredibly successful"
> — Ploum, Belgian professor, author and blogger
> https://ploum.net/2026-01-22-why-no-european-google.html

Cool article, because it points at all the European inventions the current digital ecosystem builds on:

- Linux
- Git
- HTML
- WWW
- MP3
- VLC

All made available for free. “Europeans dream big: they hope that their work will benefit humankind as a whole!”

For Friday – a company that start with [.]

1. Submit an article, blog or news item that relates to todays class
   - about a listed company that starts with the same letter as your first name

2. Put in a note (comment, annotation) where you draw on todays lecture to explain the value of the company to
   a) Investors
   b) Society

- we'll talk about this on Friday.

The Future of capitalism

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